

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V) Data
Visualization with Power BI and Tableau

Practical – V

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Class: TYIT	Batch: 1
Date of Assignment: 20-08-26	Date/Time of Submission: 20-08-26

Part B – Program

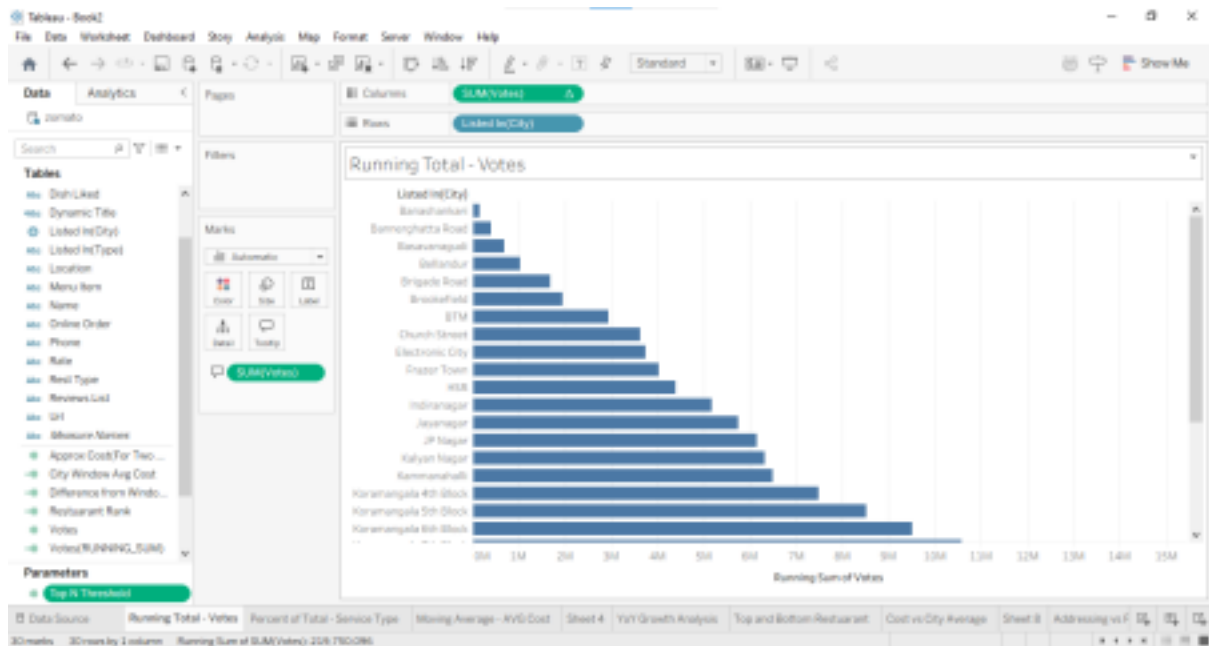
Analyzing Data using Table Calculations in Tableau

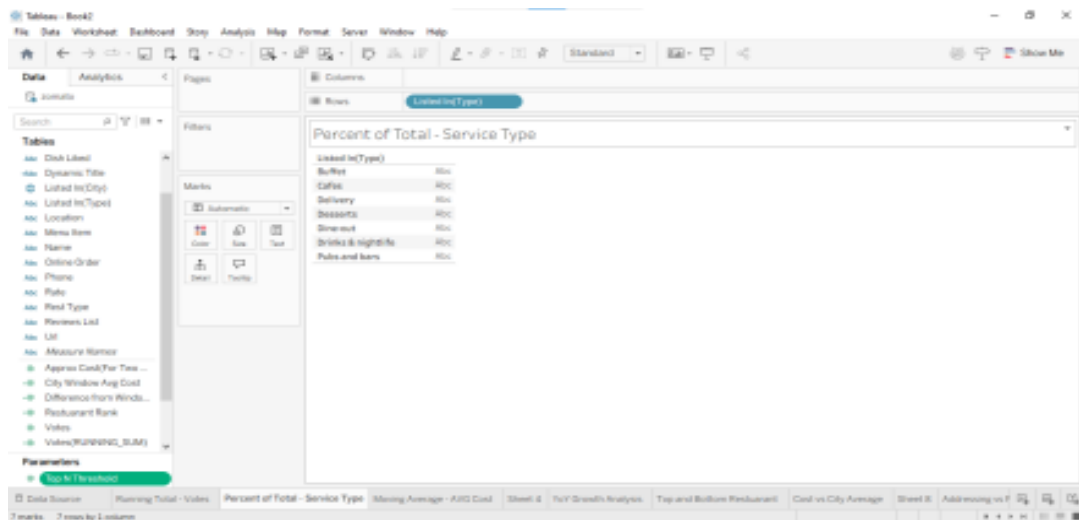
a) Running Total and Percent of Total

Steps:

1. Open zomato.csv in Tableau.
2. For Running Total: listed_in(city) → Rows, votes → Columns.
3. Right-click SUM(votes) → Quick Table Calculation → Running Total.
4. For Percent of Total: listed_in(type) → Rows, votes → Columns.
5. Select Quick Table Calculation → Percent of Total.

Output:





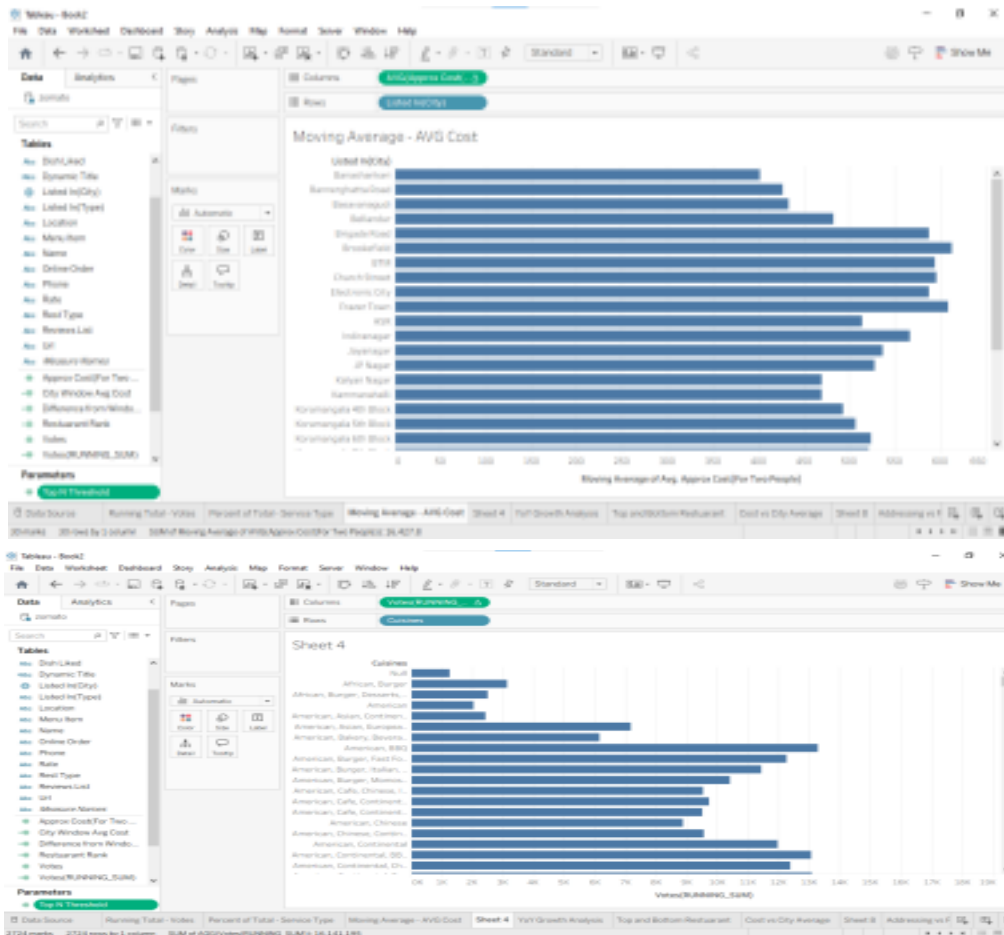
b) Create moving averages and cumulative metrics

b) Moving Average and Cumulative Metrics

Steps:

1. `listed_in(city)` → Rows.
2. `approx_cost`(for two people) → Columns and set it to **Average**.
3. Select Quick Table Calculation → Moving Average.
4. Set **Previous Values** = 2 and **Next Values** = 0.
5. Create **Cumulative Votes** using:
`RUNNING_SUM(SUM([votes]))`

Output: Moving Average of Cost and Cumulative Votes.



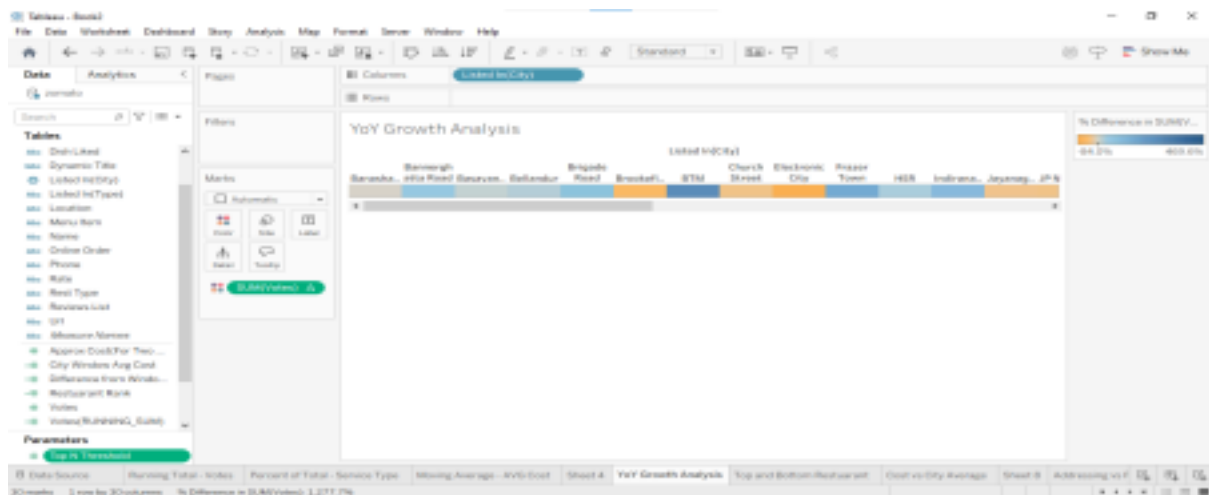
c) Perform year-over-year growth analysis

c) Year-over-Year Growth Analysis

Steps:

1. Add the time/ordinal dimension → **Columns**.
2. Add votes → Rows.
3. Select Quick Table Calculation → Percent Difference.
4. Select Compute Using → Table (Across).
5. Add the measure to **Color**.

Output: YoY / Period-over-Period Growth Analysis.



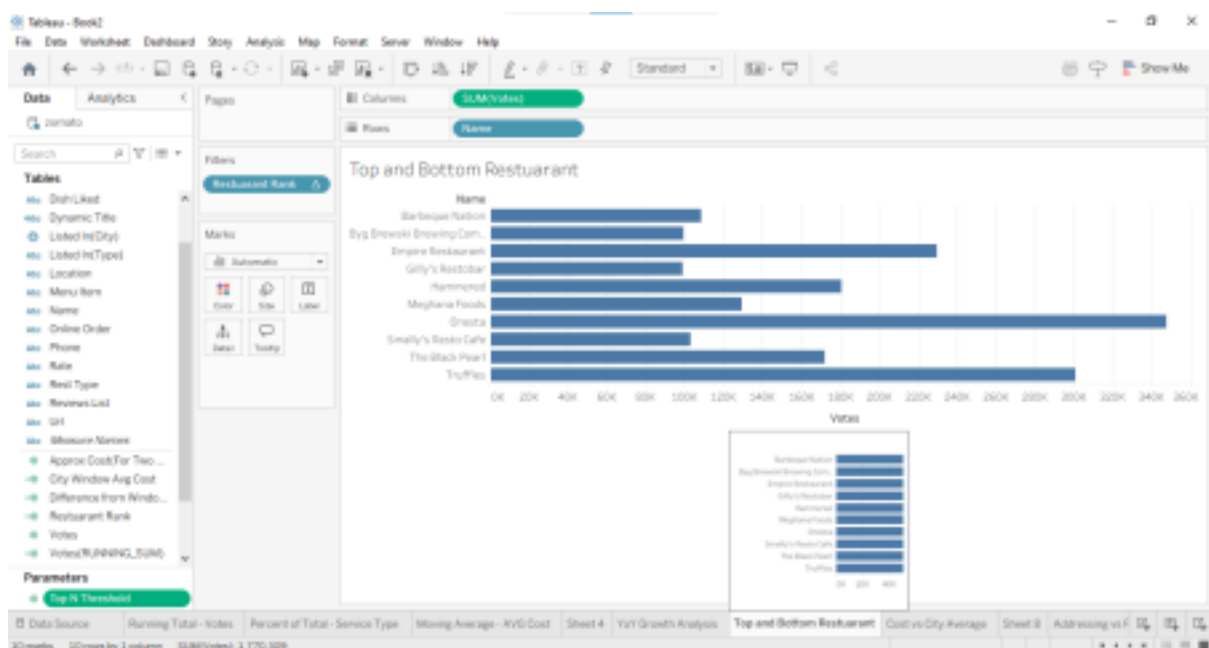
d) Use ranking functions (Top N, Bottom N)

d) Ranking Functions – Top N / Bottom N

Steps:

1. Create **Restaurant Rank**:
RANK(SUM([votes]), 'desc')
2. Add name → Rows and votes → Columns.
3. Add Restaurant Rank → Rows.
4. Drag Restaurant Rank to **Filters**.
5. Set the maximum rank to **10**.

Output: Top 10 Restaurants by Votes.



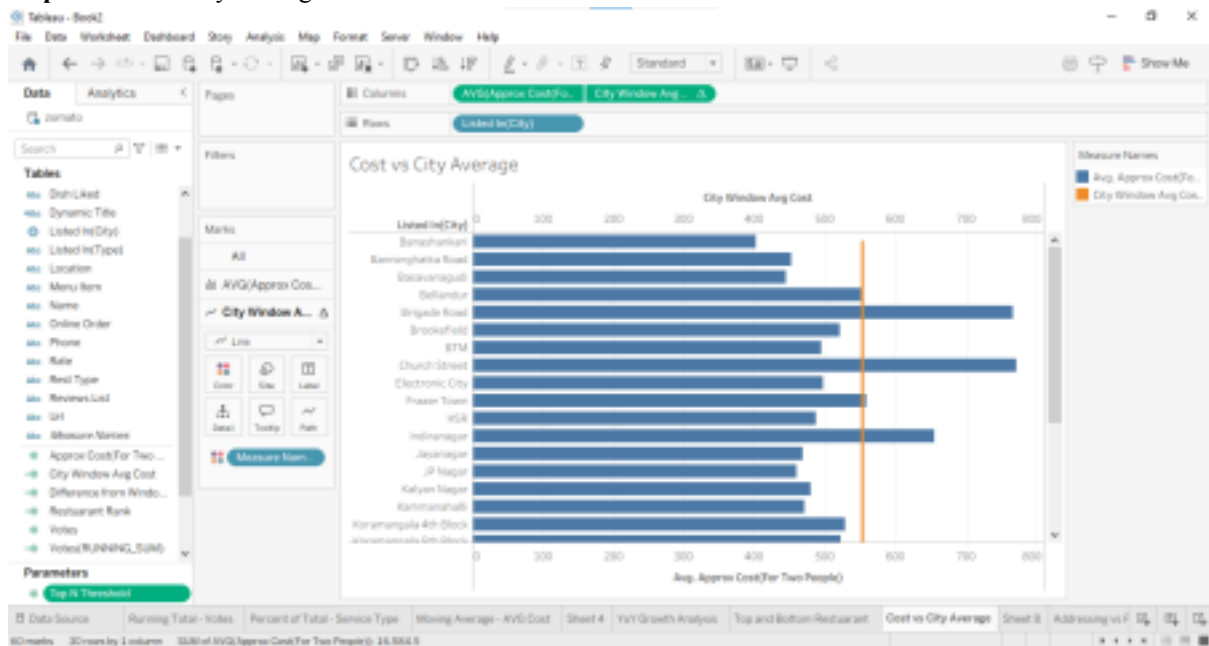
e) Implement window functions (WINDOW_SUM, WINDOW_AVG)

e) Window Functions

Steps:

1. Create **City Window Avg Cost**:
WINDOW_AVG(AVG([approx_cost(for two people)]))
2. Add listed_in(city) → Rows.
3. Add Average Cost and Window Average → Columns.
4. Select **Dual Axis** → **Synchronize Axis**.
5. Set Cost as **Bar** and Window Average as **Line**.

Output: Cost vs City Average Chart.



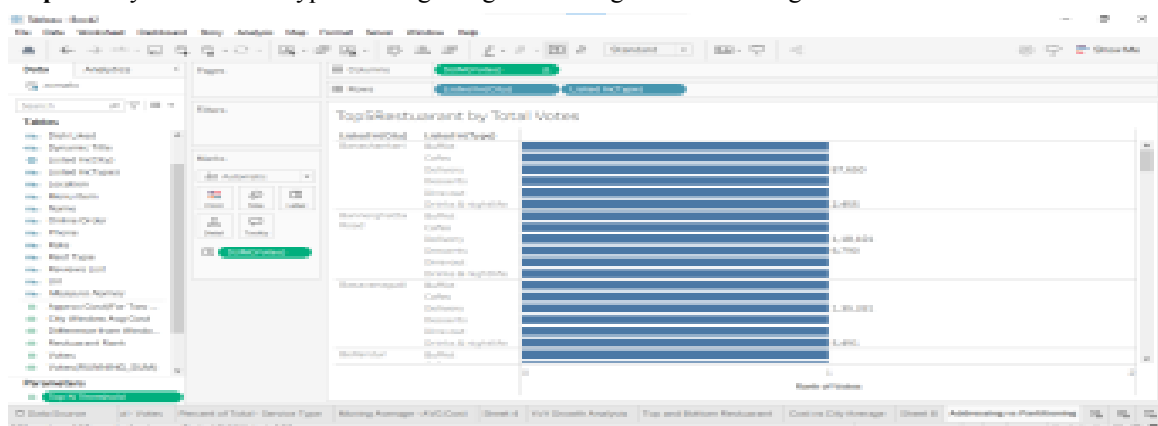
f) Customize addressing and partitioning

f) Addressing and Partitioning

Steps:

1. Add listed_in(city) and listed_in(type) → Rows.
2. Add votes → Columns.
3. Select Quick Table Calculation → Rank.
4. Use **Compute Using** → **Table (Down)** or **Pane (Down)**.
5. Use **Specific Dimensions** to customize addressing and partitioning.

Output: City and Service Type Ranking using Addressing and Partitioning.



g) Create dynamic titles and labels using table calculations

g) Dynamic Titles and Labels

Steps:

1. Create **Top N Threshold** parameter.
2. Set Integer, Current Value = **5**, Range = **3–20**.
3. Create Dynamic Title:
"Top " + STR([Top N Threshold]) + " Restaurants by Total Votes"
4. Insert the parameter into the worksheet title.
5. Add SUM(votes) → Label and select **Min/Max**.

Output: Dynamic Top-N Title and Dynamic Mark Labels.

